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Sri Lanka Institute of Information Technology

# B.Sc. Special Honours Degree
in
# Computer Science

Test 01 -Model Paper
Year 2, Semester 2 (2024)

# Design and Analysis of Algorithms (SE2022)

Duration: 1 Hour

| | |
|---|---|
| Q1 | |
| Q2 | |
| Total | |

Instruction to Candidates:

- This paper has 2 questions. Answer All Questions.
- Total Marks 20.
- This paper contains 7 pages with Cover Pager.
- Calculators are allowed.

**QUESTION ONE ( 10 Marks )**

(1) List two properties of good algorithm. (02 marks)

(2) What is the main purpose of analysis of algorithms? (02 marks)

(3) Analyze the running time of the following program fragment assuming a RAM model of computation. (03 marks)

```
j ← 0
for i ← 0 to n
    while j ≤ 5
        j ← j +1
```

(4) What are the best case and worst case running time (in Big O notation) of the insertion sorting algorithm? When do you get such situations in the insertion sorting algorithm? (02 marks)

(5) Illustrate the operations of the *Insertion sort algorithm* for the array with the given set of elements. (For the illustration process assign the values only once to the given algorithm codes and then use diagrammatic way to reach the answer.) (03 marks)

A

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 10 | 0 | 8 | 4 |

```
INSERTION-SORT(A)
1 for j ←2 to length[A]
2    do key ← A[j]
3        ▷ Insert A[j] into the sorted sequence A[1..j-1]
4        i ← j - 1
5        While i > 0 and A[i] > key
6                do  A[i+1] ← A[i]
7                    i ←i-1
8         A[i+1] ← key
```

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(6) Find the *Big oh* value for the following functions. Justify your answer. (02 marks)

a) $T(n) = \log n + n + 100$

b) $T(n) = 200n^{20} + 100n^{10} + 2^n + 3^n + n + 1$

(7) The power function can be defined as $pow(x, n) = x^n$. This can be evaluated using the multiplication as $x^n = x \times x^{n-1}$ where $x$ is any real number and $n$ is a non-negative integer. [Hint: $pow(x, n-1) = x^{n-1}$] (06 marks)

a) Write a recursive relation for $pow(x, n)$ where $x$ is any real number and $n$ is a non-negative integer. Clearly define the initial condition(s).
b) Write a recursive algorithm in pseudo code for the above recursive relation.
c) Write a recurrence equation that describe the running time T ($n$) for the above part b) recursive algorithm.

**QUESTION TWO (10 marks )**

(1) Given below is an algorithm for *QUICKSORT* (*A, p, r*)

```
Procedure QUICKSORT (A, p, r)
 1  if p < r
 2    then q ← PARTITION(A, p, r)
 3         QUICKSORT (A, p, q-1)
 4         QUICKSORT (A, q+1, r)

Procedure PARTITION(A, p, r)
1   x ← A[r]
2   i ← p - 1
3     for j ← p to r - 1
4        do if A[j] ≤ x
5              then i ← i + 1
6                   exchange A[i] ↔ A[j]
7     exchange A[i + 1] ↔ A[r]
8   return i + 1
```

a) Illustrate the operations of the *QUICKSORT (A, p, r)* for the array with the given set of elements. (For the illustration process assign the values only once to the given algorithm codes and then use diagrammatic way to reach the answer.) (03 marks)

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| 28 | 15 | 1 | 30 | 0 | 8 |

b) Depending on the way array is partitioned by the quick sort algorithm, best case or worst case can occur. What kinds of partitioning generate the best or worst case?
(02 marks)

c) Write recurrence equations for best case in the above part b).

d) Use the repeated substitution method to solve the best case recurrence equation.
(5 marks)

~~~~~~~~~~~~~~~~END OF THE PAPER~~~~~~~~~~~~~~~~
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